

UNDERSTANDING GLOBAL WARMING

HUMAN ACTIVITY IS ENHANCING THE EARTH'S GREENHOUSE EFFECT AND WARMING ITS ATMOSPHERE

Over the past 200 years, Earth's average temperature has risen rapidly against a prevailing trend of cooling. Evidence suggests that this global warming is not the result of natural climate variation but is due largely to human activity—and it could have disastrous consequences.

Energy from the Sun reaches Earth in the form of electromagnetic radiation—mostly visible light and infrared and ultraviolet radiation. Some of the energy is absorbed, with the rest reflected back into space. The absorbed energy heats the planet—and since any warm object emits infrared radiation, Earth also loses heat to space. At a certain temperature, the planet radiates energy at the same rate as it absorbs it.

GREENHOUSE EFFECT

With no atmosphere, Earth's equilibrium temperature would be around -1°F (-18°C). However, the atmosphere absorbs some of the incoming and outgoing radiation, and it warms up. The warmed atmosphere produces its own infrared radiation, some of which is absorbed by

the surface below. As a result, the equilibrium temperature is higher—around 59°F (14°C). This phenomenon is known as the greenhouse effect, since a greenhouse also “traps” heat, making it warmer than it would otherwise be.

GREENHOUSE GASES

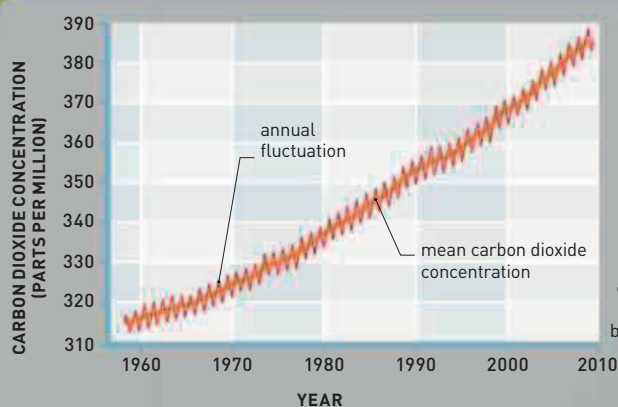
The first experimental evidence for the greenhouse effect came from Irish physicist John Tyndall. In the 1850s, Tyndall measured how much infrared radiation various gases absorb. The strongest “greenhouse gas” is water vapor, but methane, carbon dioxide, and ozone also contribute significantly. In the 20th century, climate scientists found that the concentration of carbon dioxide is increasing, enhancing the greenhouse effect and raising the equilibrium



JOHN TYNDALL
Irish physicist John Tyndall (1820–93) studied magnetism and atmospheric physics. He was a great popularizer of science.

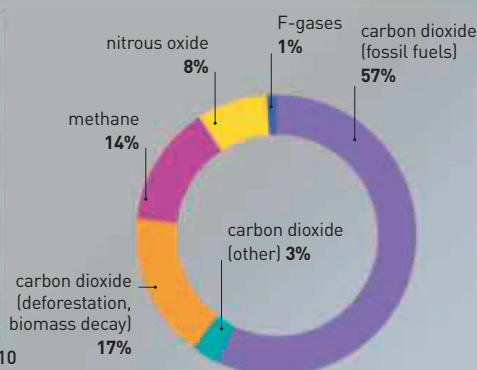
“GLOBAL WARMING MUST BE SEEN AS AN **ECONOMIC** AND **SECURITY** THREAT.”

Kofi Annan, former United Nations Secretary General, 2009



Keeling curve

In 1958, American scientist Charles Keeling began a program to monitor atmospheric carbon dioxide. The concentration has now reached 400 parts per million, compared with 280 before the Industrial Revolution.



Emissions by greenhouse gases

These gases are released as a result of human activity. Carbon dioxide contributes most to global warming.

total of 30 percent of incoming energy (52 PW) reflected without being absorbed

4 percent of energy reflected by atmosphere

20 percent of energy reflected by clouds

6 percent of energy reflected from Earth's surface

amount of reflection depends on land covering—snow reflects more than soil

atmosphere and clouds